

The Upcycled Campus

Student Housing in Rotterdam Campus

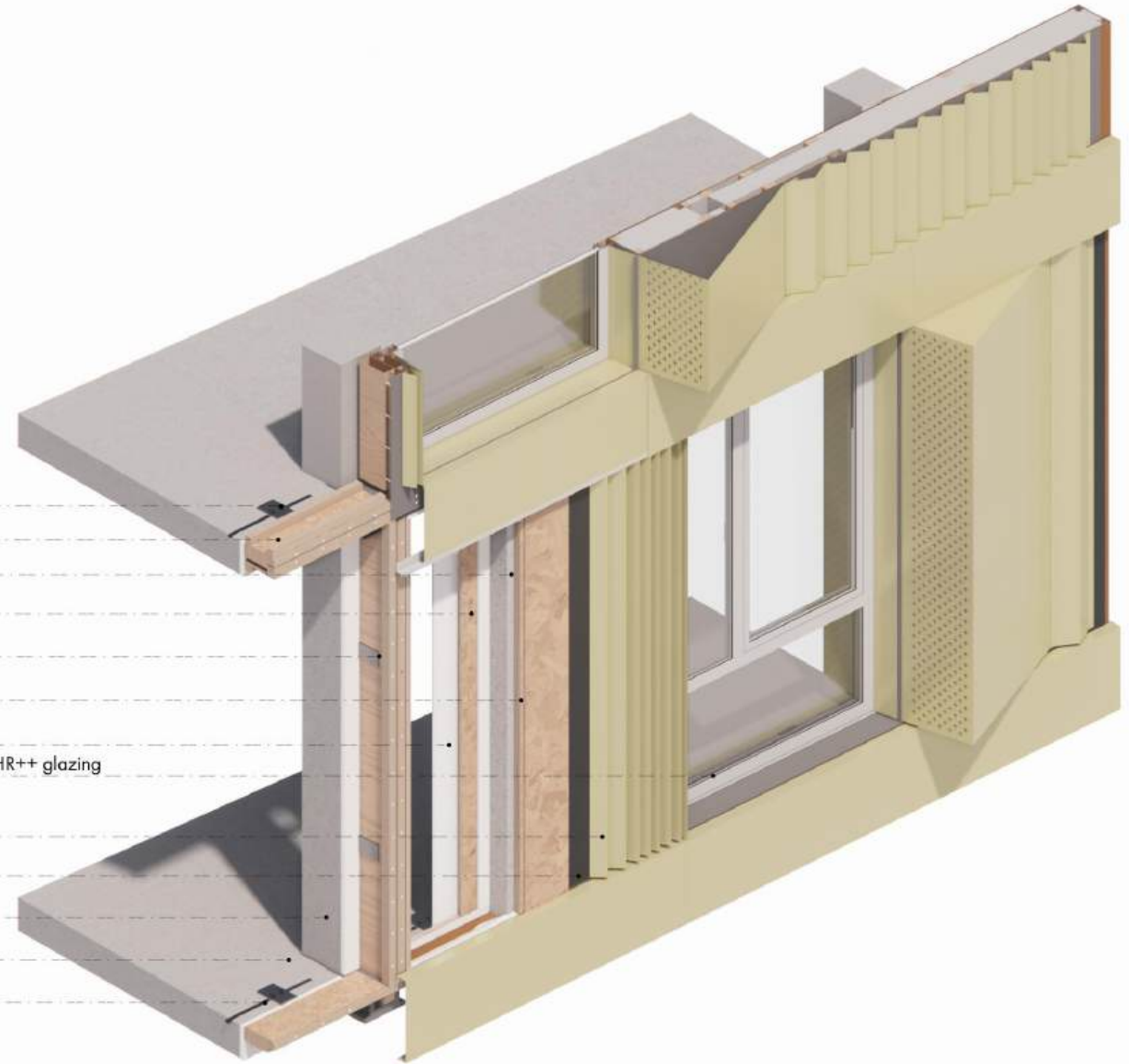




Case study part-A

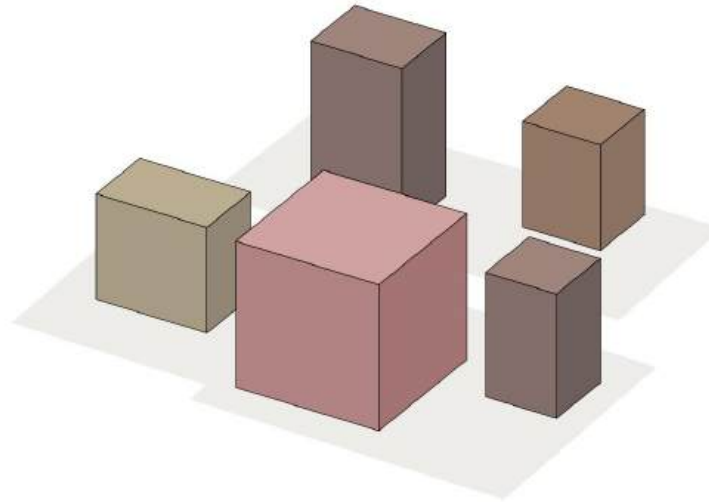
Erasmus Campus student housing

Steel Z Anchor
 Timber Framework
 EPS Insulation
 Chipboard Extra WR P5
 Steel U Profile
 SIPS Stiffeners
 Fermacell Board
 Window frames Reynaers CS 77 RAL9004 with HR++ glazing
 Alucoil Larson A2 Aluminum Panel
 Water Resistant MergoFassade UV-FR
 CD20 Prefabricated concrete column
 CD20 Prestressed concrete slab
 U-Channel Railing Profile

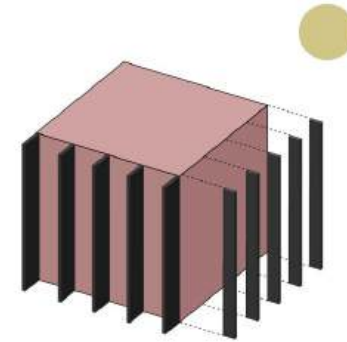


Concept

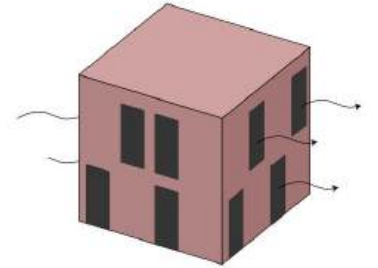
- Student housing in Erasmus Campus in Rotterdam
- User-friendly Facade



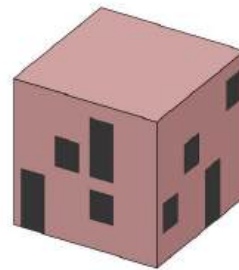
Harmonious Integration



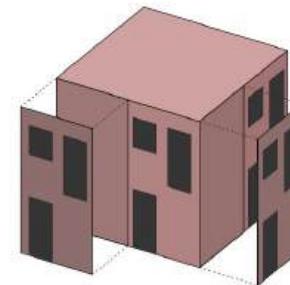
Shading & Cooling Solutions



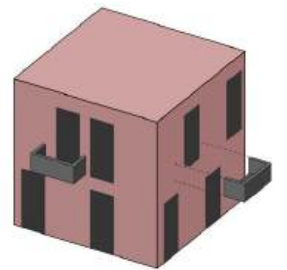
Ventilation & Comfort



Privacy



Modularity for flexibility



Balconies

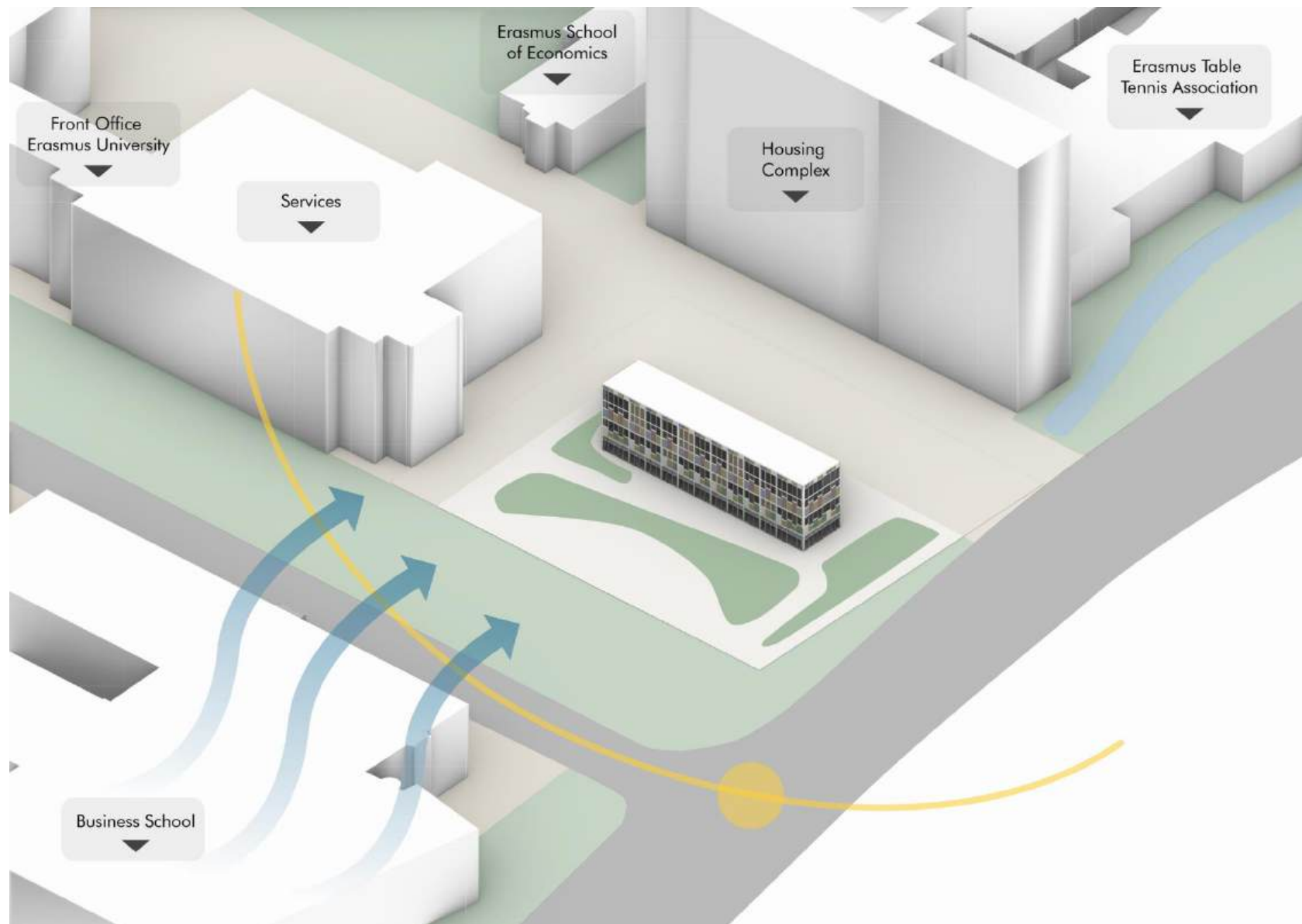
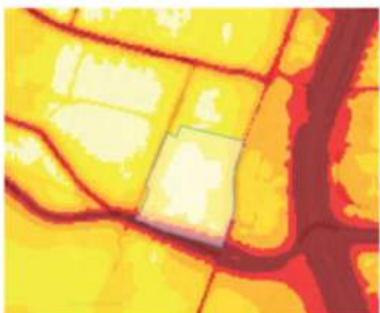
Climate and Urban context

Map Location



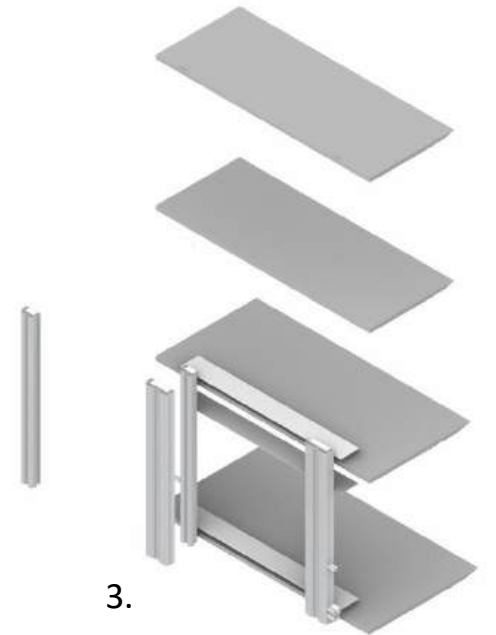
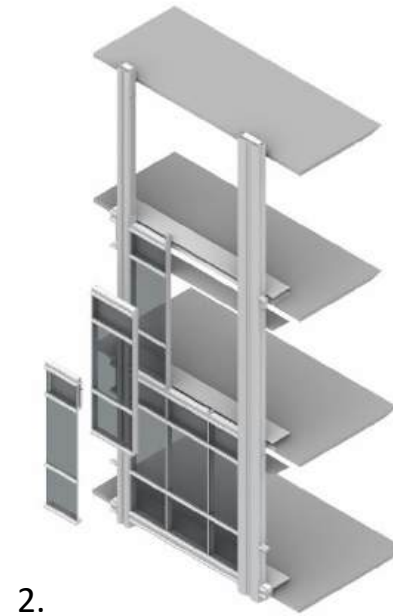
Rotterdam Erasmus Campus
P4, 3062 PA Rotterdam

Environmental Noise





Canary Warf City Bank



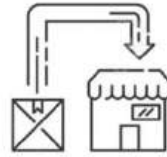
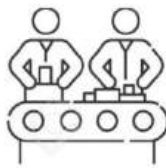
Steps to Disassembly

Design Criteria

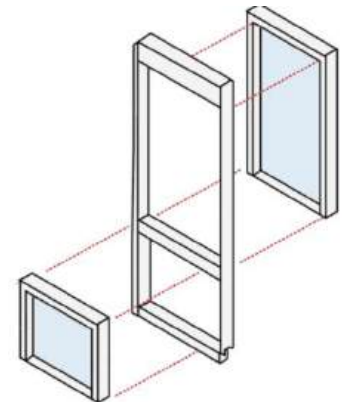
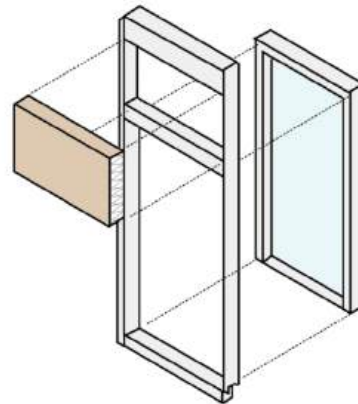
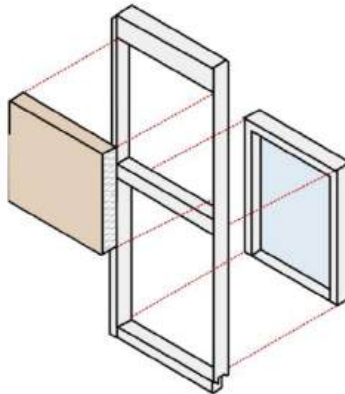
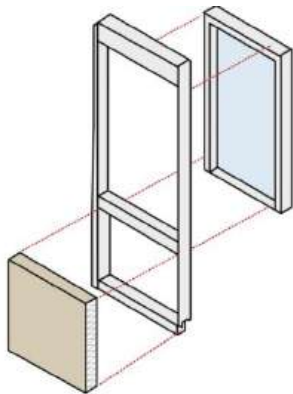
Maximise reuse



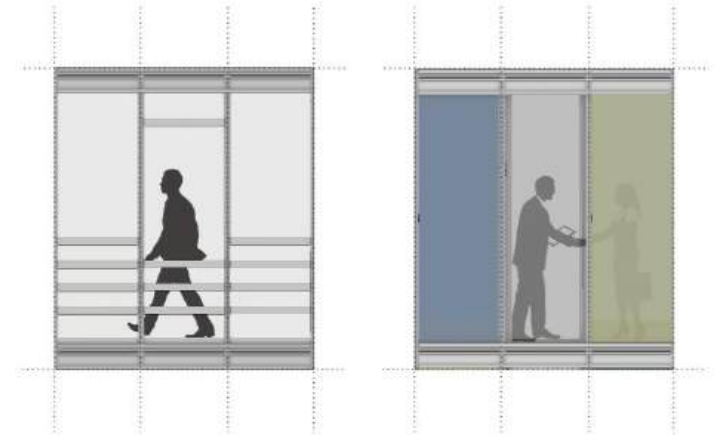
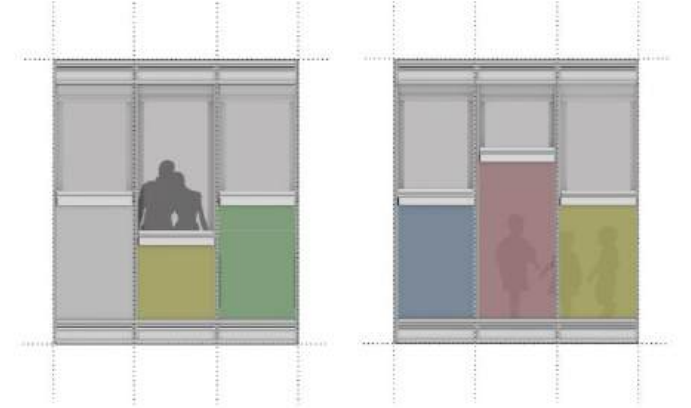
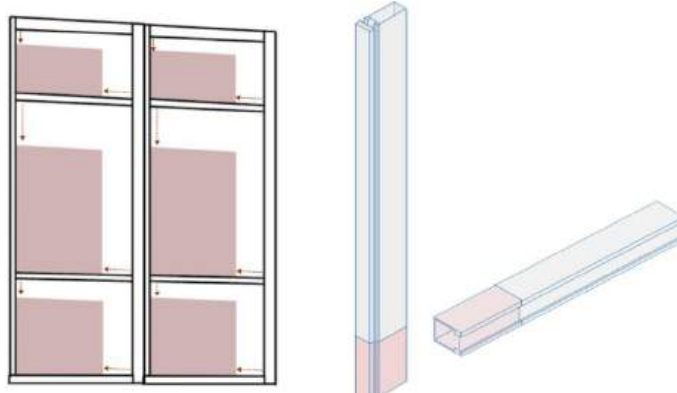
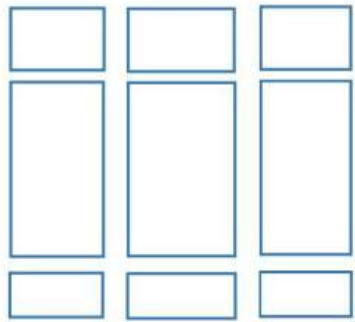
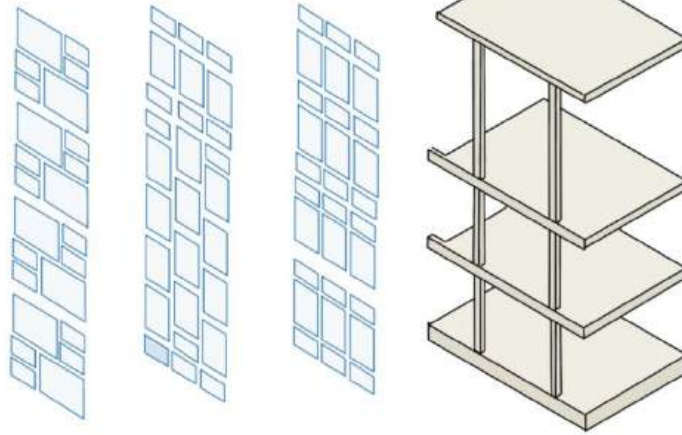
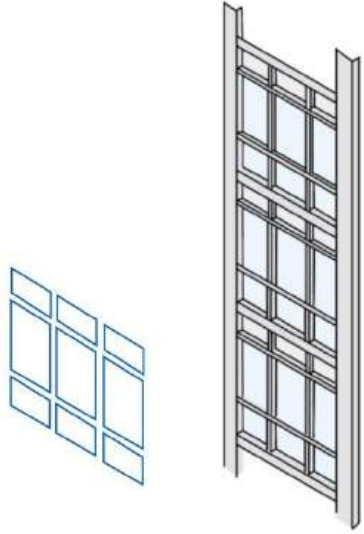
Ease and speed of
assembly and
production



Versatility



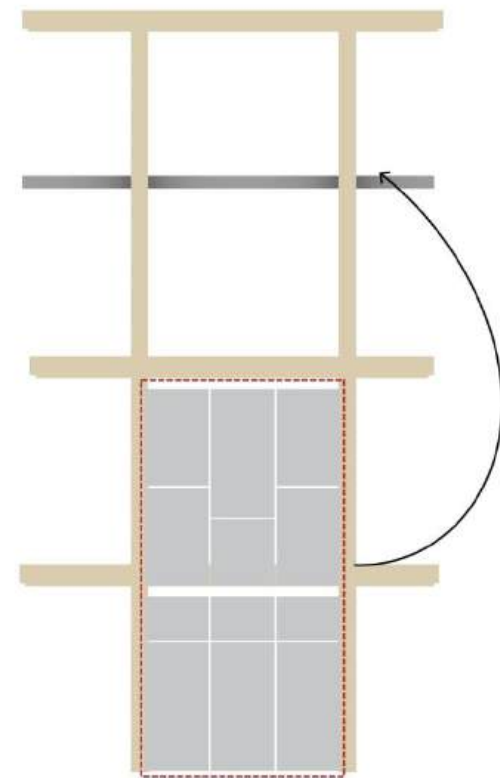
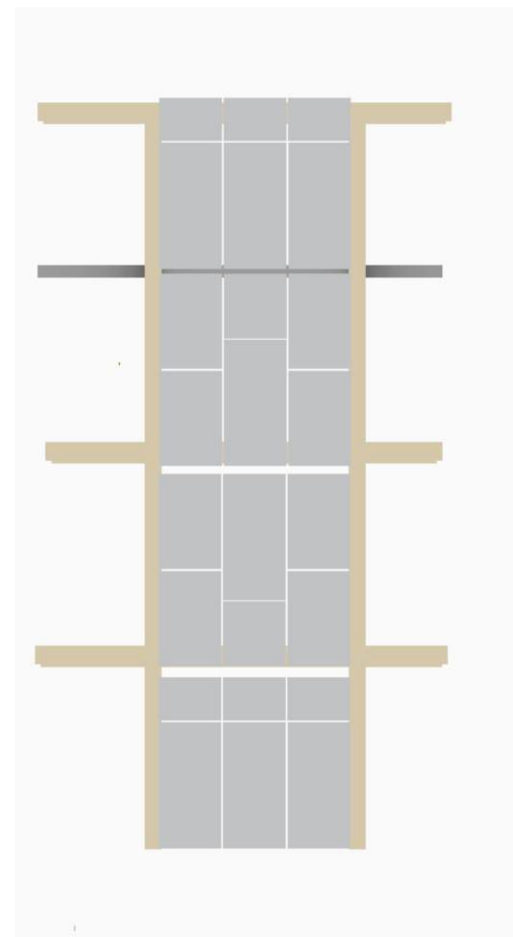
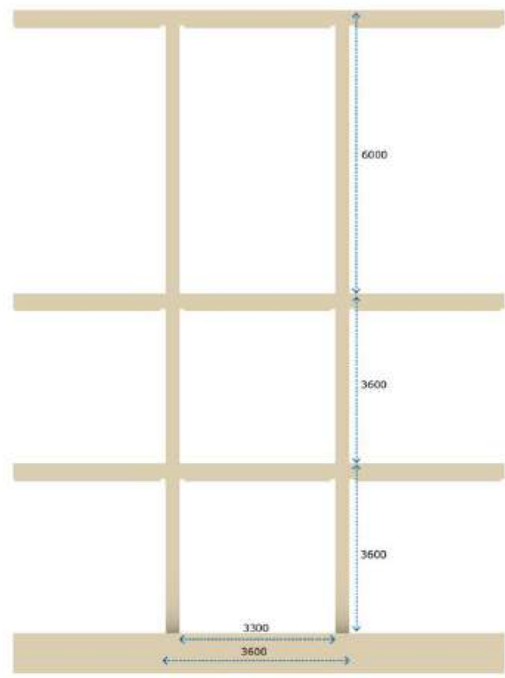
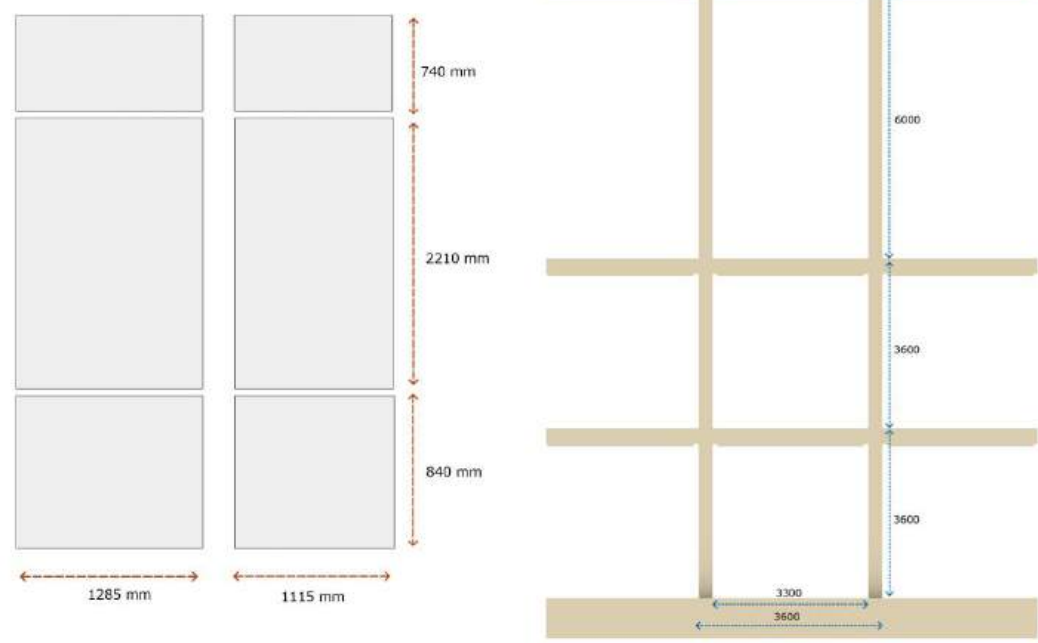
Design Concept



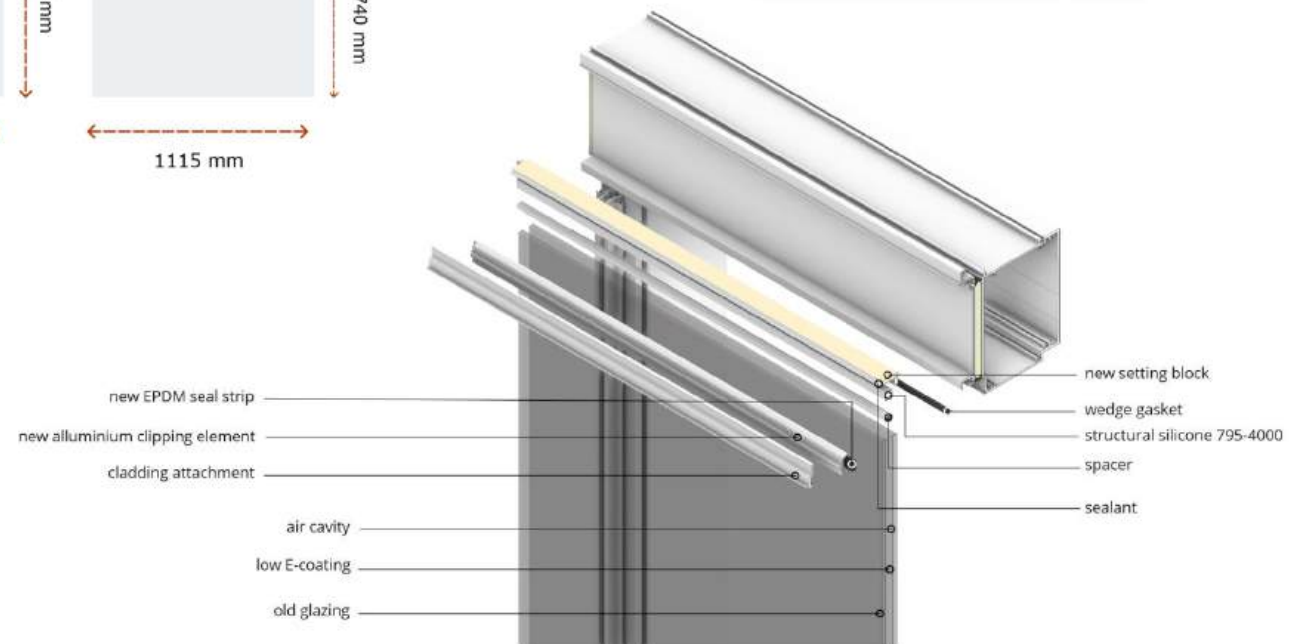
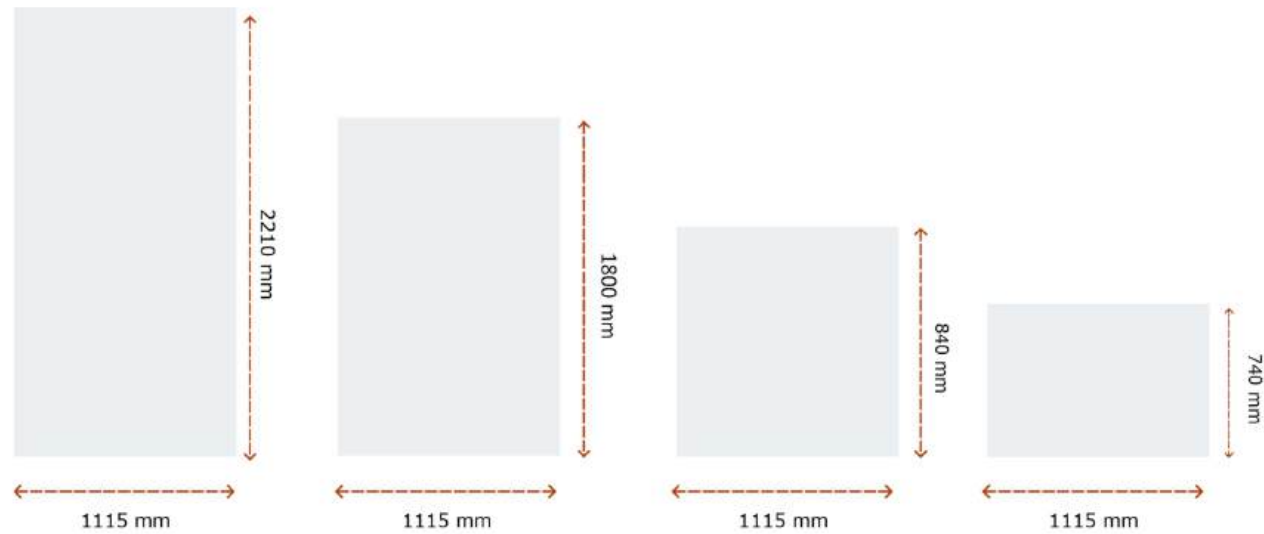
Reflections and transparency



Approach Development



Glazing



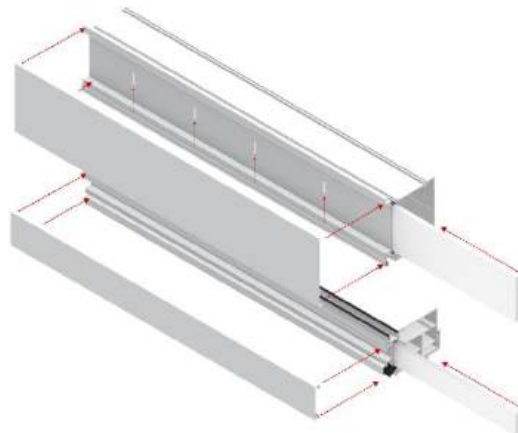
Mullions Remanufacturing



old cladding - to be flattened

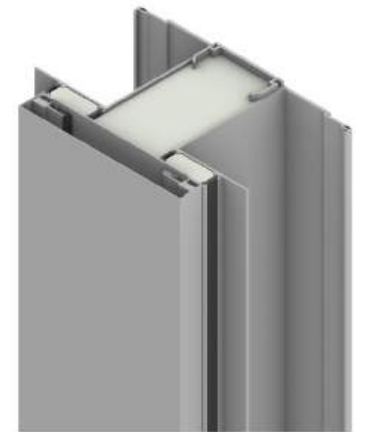
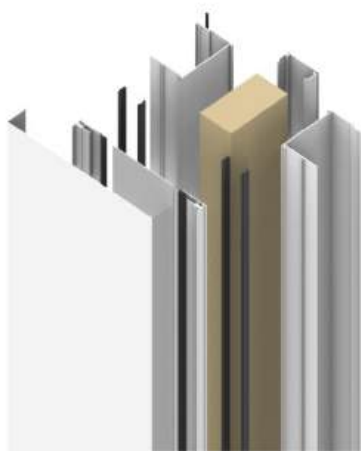
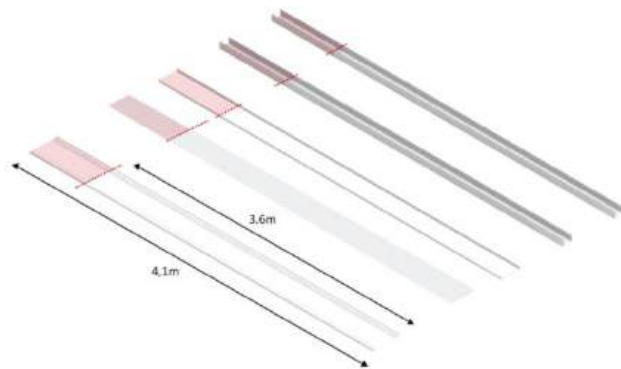
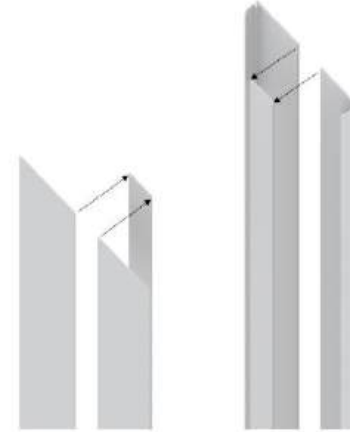
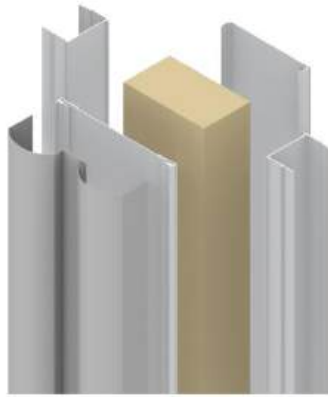


aluminium profile cutting

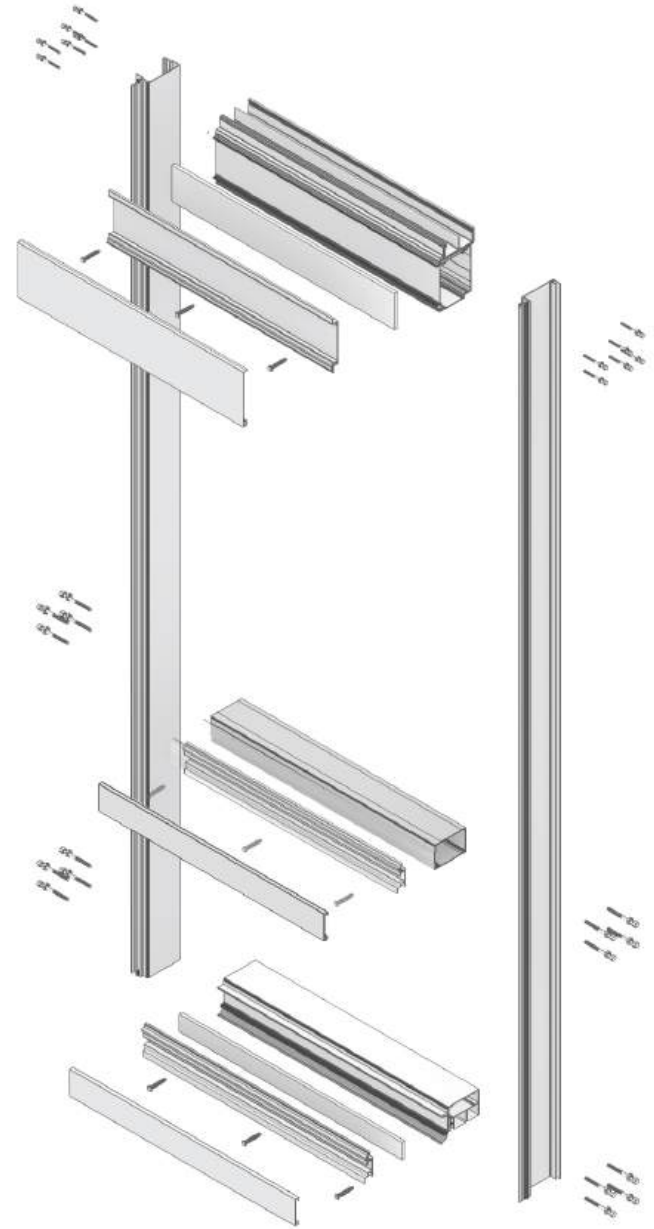
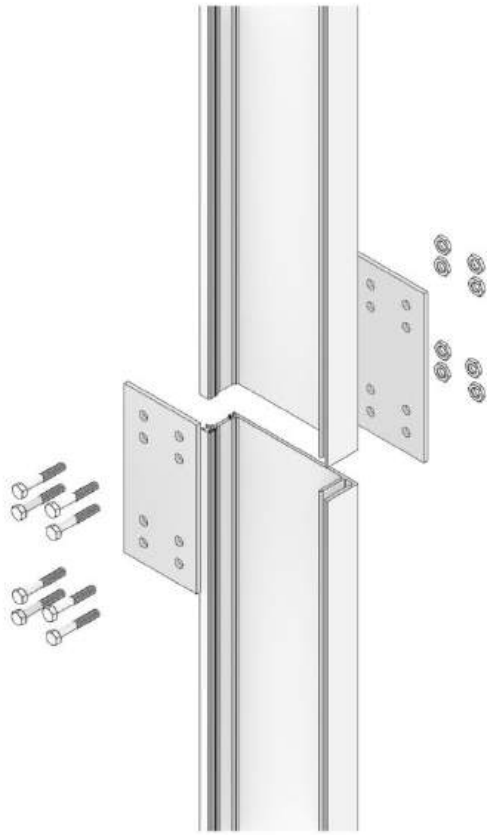


aluminium sheet flattening

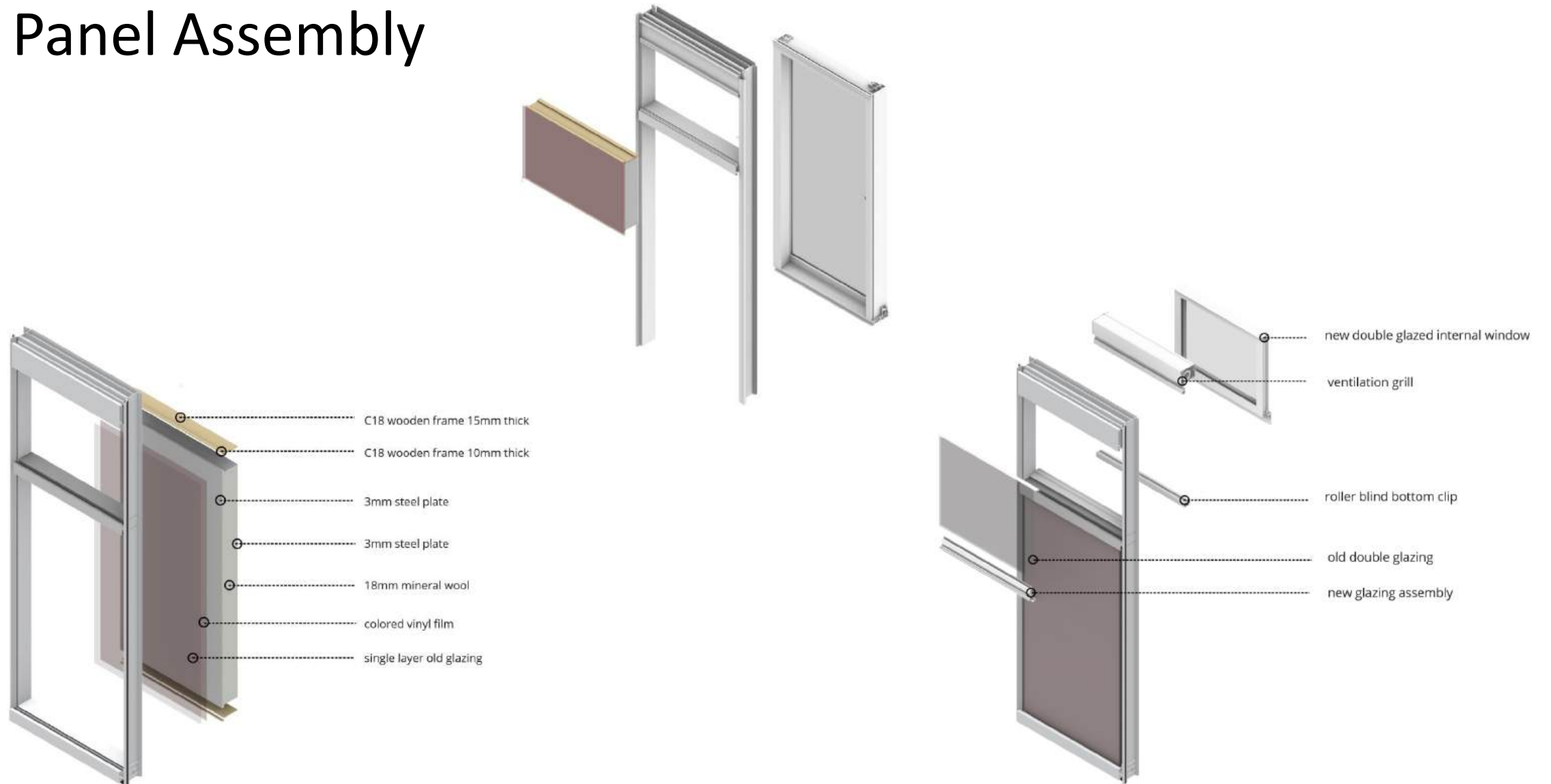
Transoms Remanufacturing



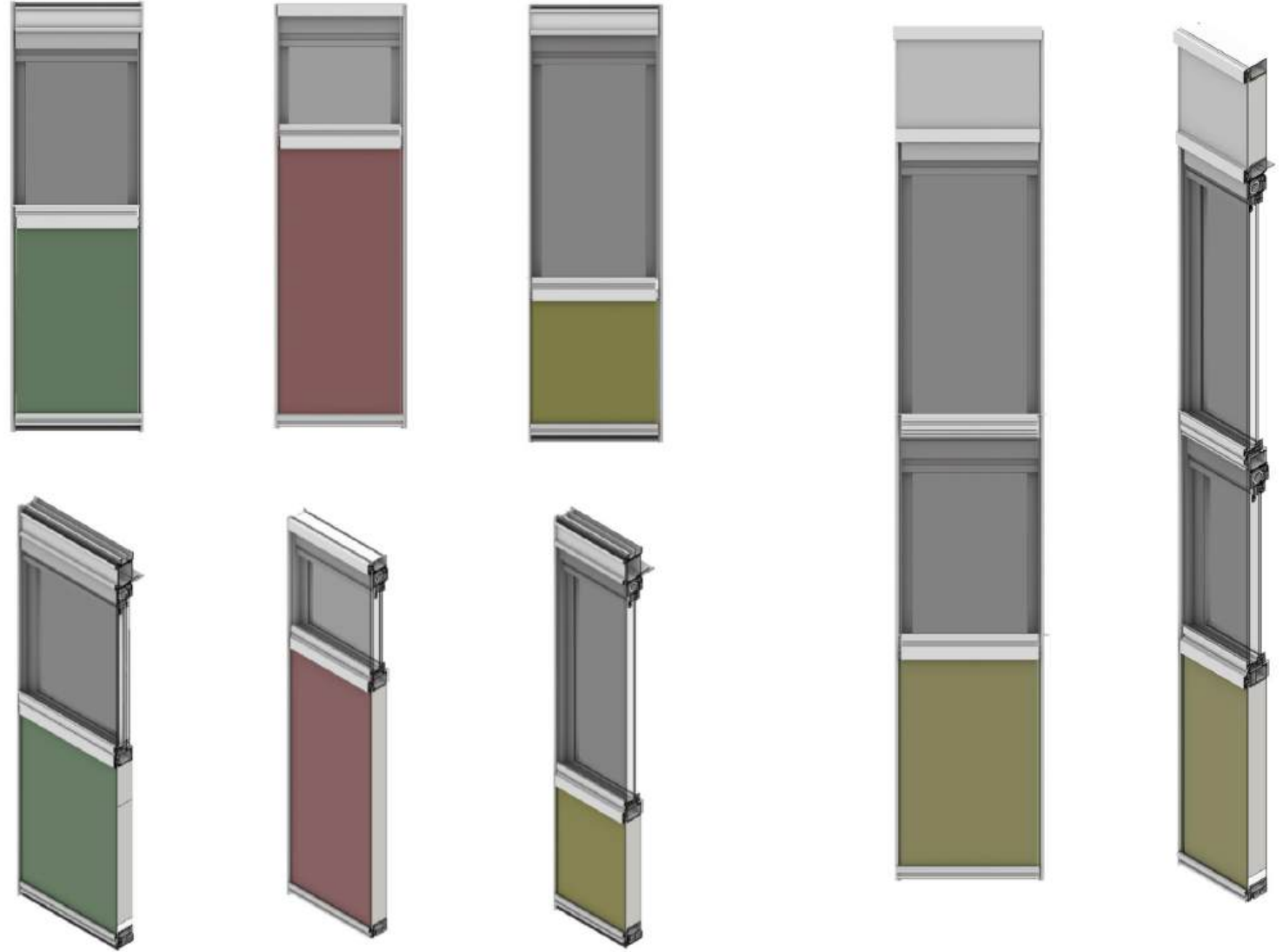
Panel Assembly



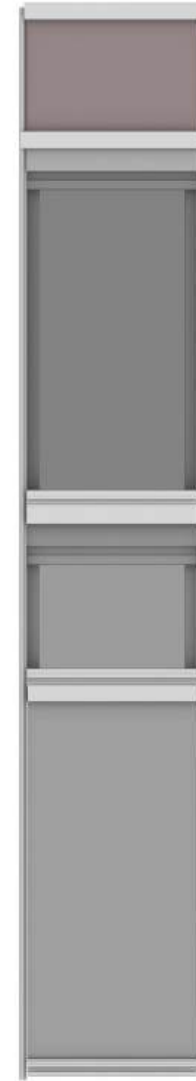
Panel Assembly



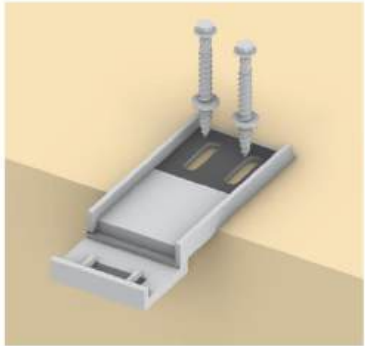
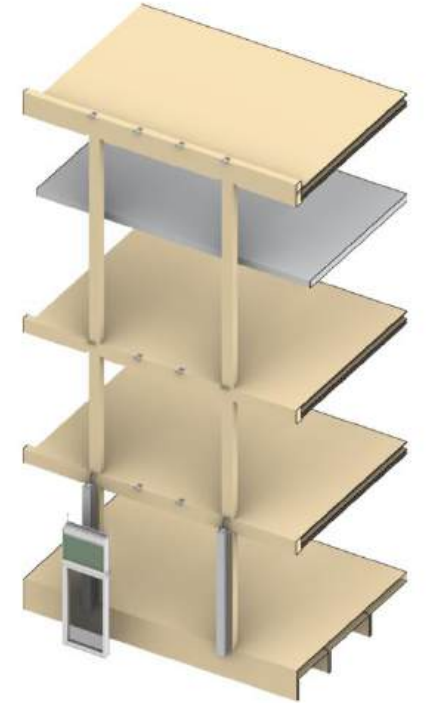
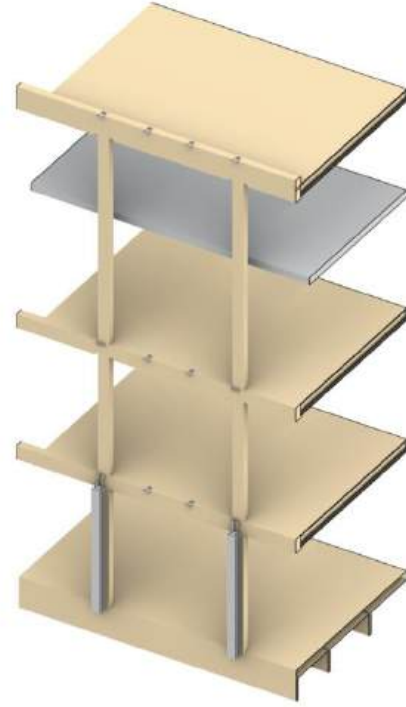
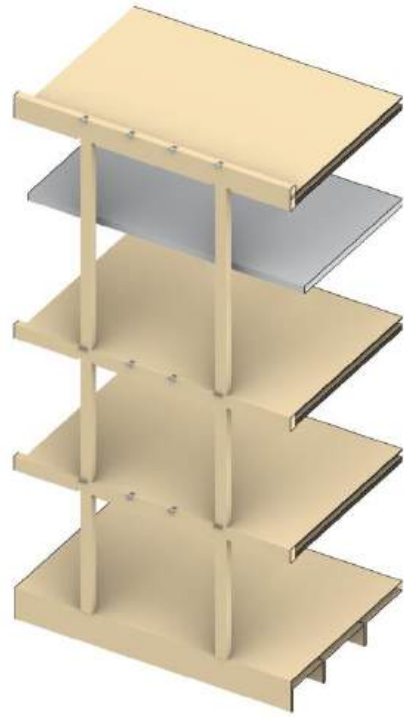
Panel type overview



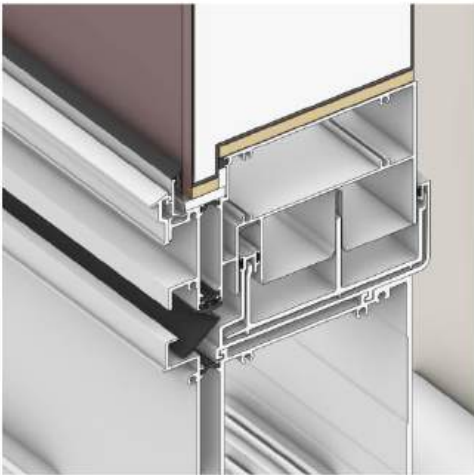
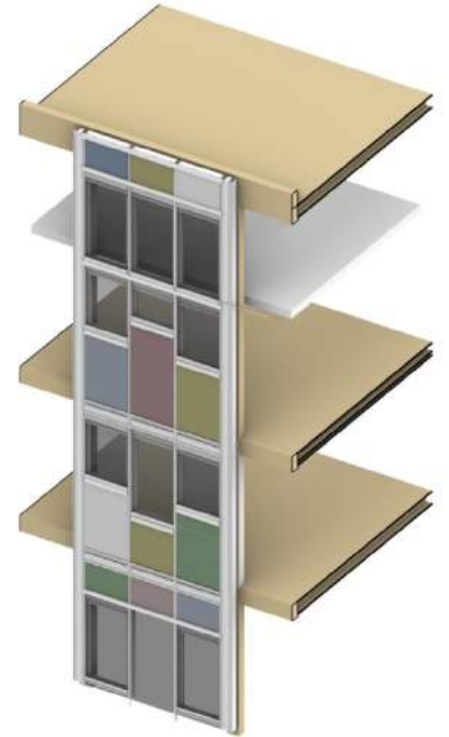
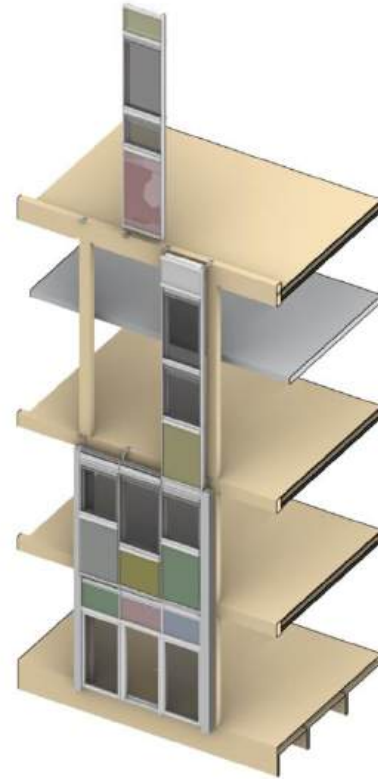
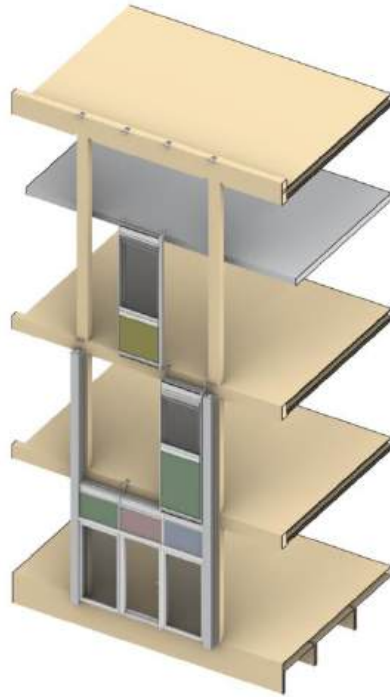
Panel type overview



On-site assembly



On-site assembly



- Transoms of the Curtain wall frame

- FIXVENT MONO AK EVO Ventilation grill

- Polyester blinds

- PLANICLEAR Glass

- SGG DIAMANT extra clear glass

- THM 90 EVO Ventilation grill

- New Aluminum cladding

- PLANICLEAR Glass

- Self adhesive foil plane colored window film

- Aluminum Cover Panel

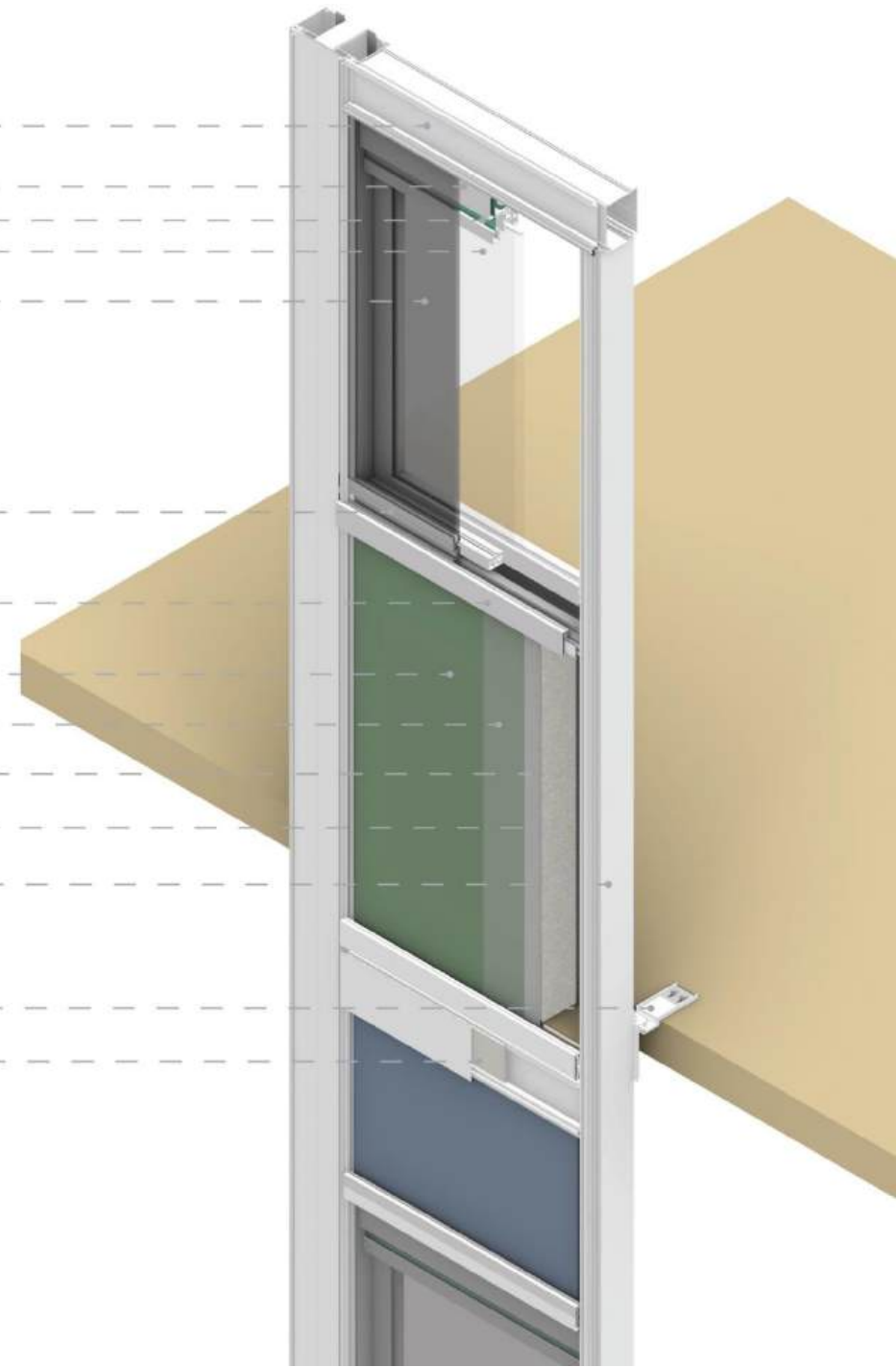
- Unidek Platinum based insulation board

- Mullions of the Curtain wall frame and new cladding

- New Steel anchors

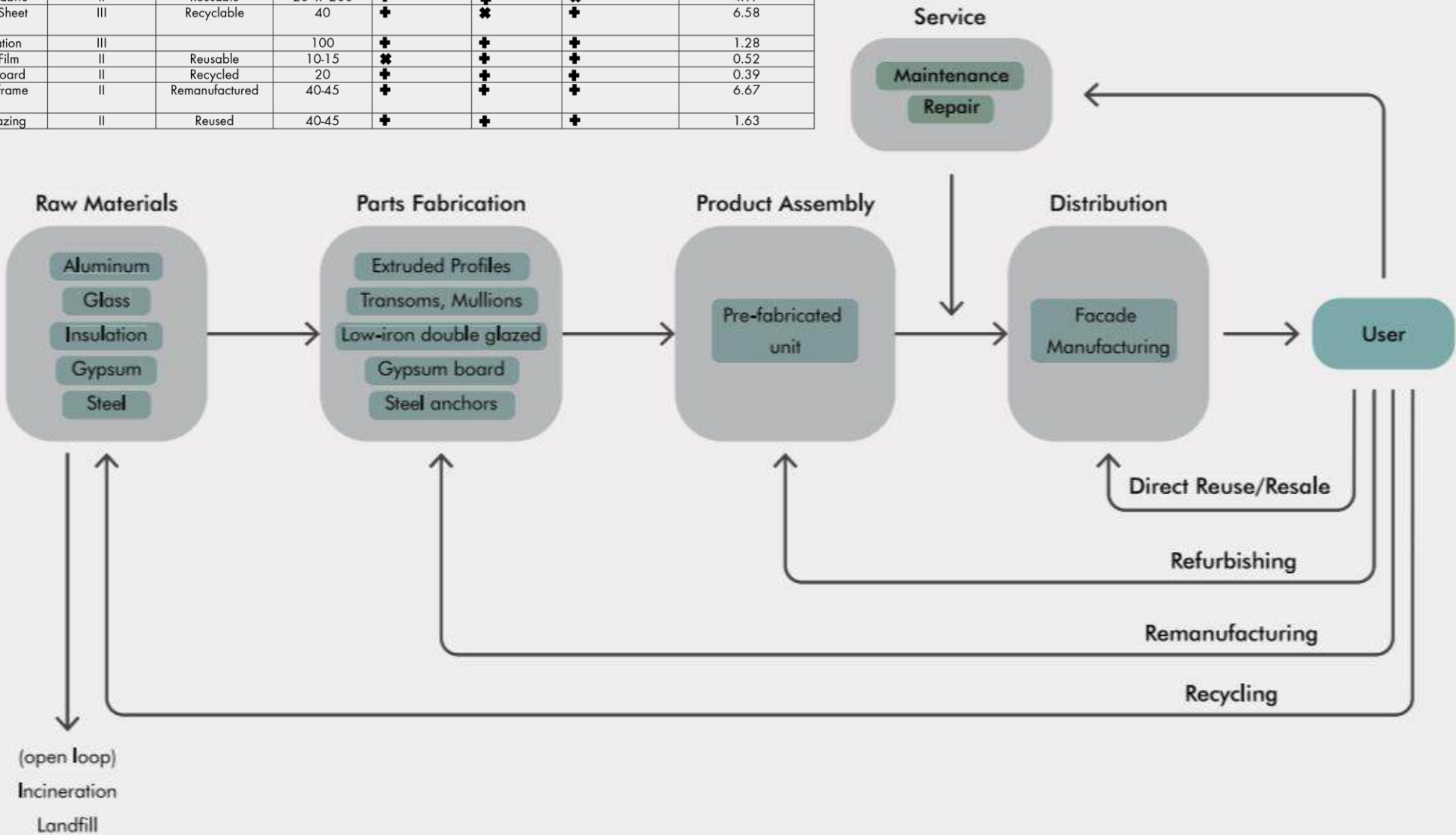
- Alkreflex Insulation layer

Elements- Materials



Product	Material	Disassembly	End of Life	Life Expectancy (Years)	Energy Efficient	Adaptability	Waste Reduction	Embodied Carbon KgCO2e/kg
Curtain Wall	Aluminum transoms and mullions	III	Remanufactured	50	+	+	+	6.83
	Double glazing	II	Reused	40-45	+	+	+	1.63
	Plastic (LDPE) grill	II	Recyclable	20-50	+	✖	+	2.08
	Steel Anchors	III	Recycled	50	+	+	+	1.27
New windows	Insulation	III	Recycled	80-100	+	✖	+	1.35
	Aluminum frame	II	Recyclable	40-45	+	+	+	6.67
	Double glazing	II	Reusable	40-45	+	+	+	1.63
	Polyester Fabric	II	Reusable	20<x<200	+	+	✖	4.97
New insulation layer	Aluminum Sheet	III	Recyclable	40	+	✖	+	6.58
	EPS insulation	III		100	+	+	+	1.28
	Colored Film	II	Reusable	10-15	✖	+	+	0.52
	Gypsum board	II	Recycled	20	+	+	+	0.39
New sliding doors	Aluminum frame	II	Remanufactured	40-45	+	+	+	6.67
	Double glazing	II	Reused	40-45	+	+	+	1.63

Circularity

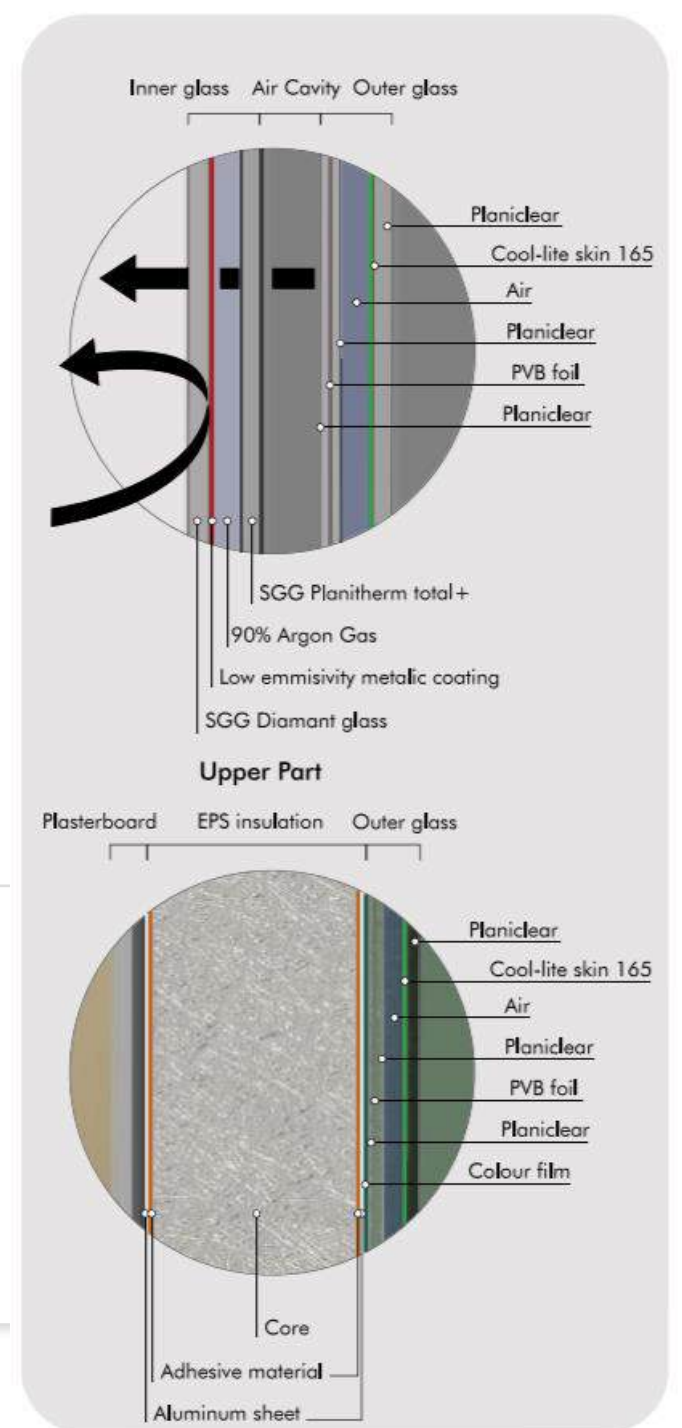


Outer Glass - old one	
U-value _(tot) (W/m²K)	Solar factor (g _{tot})
1.5	0.34

Inner Glass new one			
U-value (W/m²K)		Solar factor (g)	
Air	Argon (90%)	SGG Planiclear clear float glass	SGG DIAMANT glass low-iron
1.4	1.2	0.73	0.75
U _{tot} = 1.55 W/m²K			

EPS insulation	Plasterboard
U-value	U-value
0.21	4.64

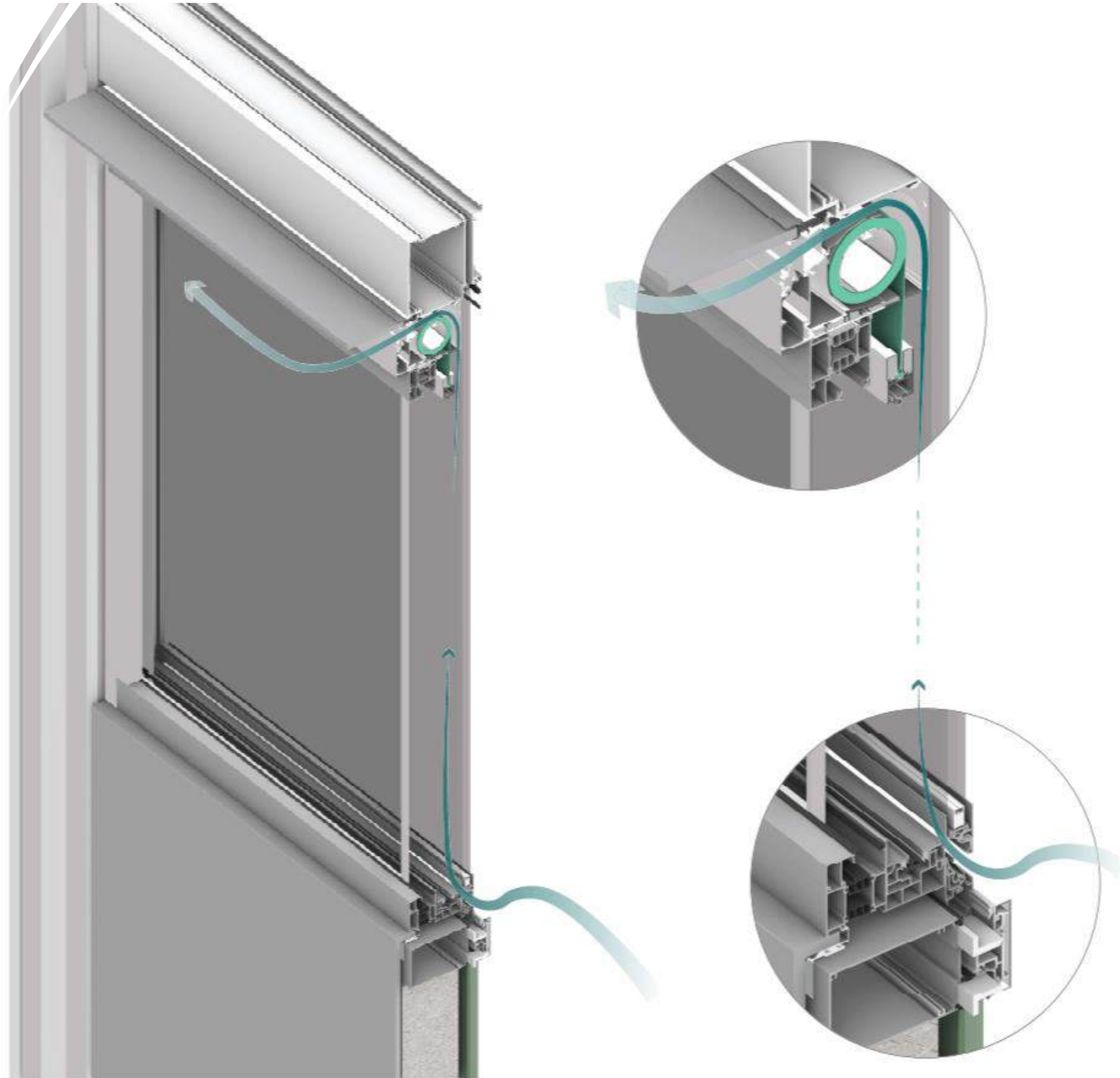
Thermal Properties

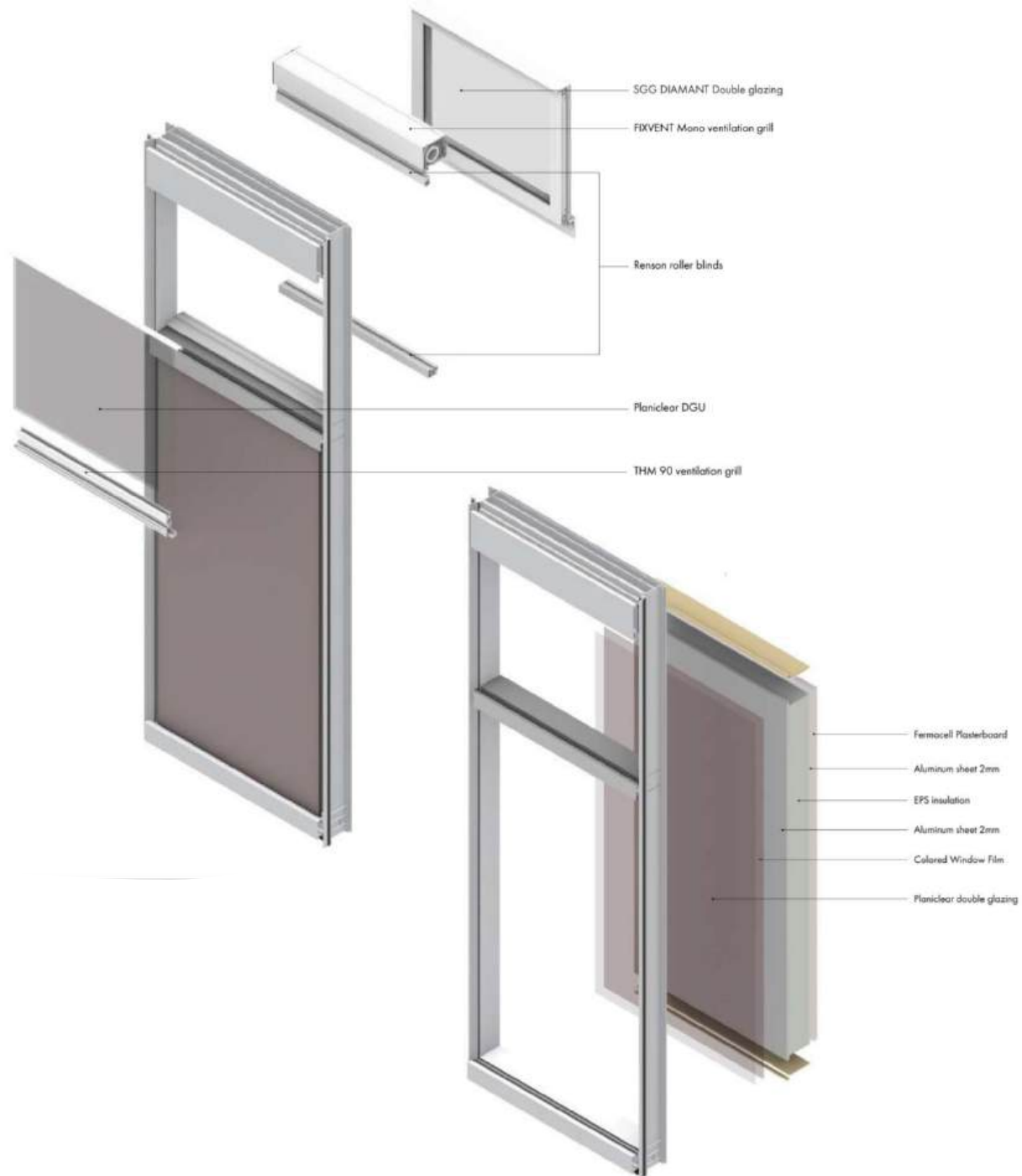


Ventilation

Upper Part	
airflow rate under 2 Pa	airflow rate under 10 Pa
9.1 m ³ /h/m	20.2 m ³ /h/m

Lower Part	
airflow rate under 2 Pa	airflow rate under 10 Pa
52 m ³ /h/m	58 m ³ /h/m





Acoustics

